

Item Code	: 204xxx draft 04 Oct 2023
PM Code	: Ins.P - PZT (AU)
Measurement(mm) ± 2mm	: 260(L) x 110(W)
Cursor (Attachment 1)	:
Colour of Artwork	: K
Source of Artwork	: MAC 1 / CC

AFT - Pyrazinamide Tablet 500mg

Qualitative and Quantitative Composition

Each tablet contains :
Pyrazinamide 500mg

Pharmaceutical Form

A white to off-white color round tablet, one side impressed with 'YSP' and the other side with a score.



Clinical Particulars

Therapeutic indications

Indicated in combination with other antituberculars, in the treatment of tuberculosis after failure with the primary medications. It is effective only against mycobacteria.

Dose and method of administration

Recommended doses by mouth for adults range from 15 mg to 35 mg per kg body-weight to a maximum of 3 g daily given in 3 or 4 equally spaced doses. Doses of 20 mg per kg have been used for children when there is no satisfactory alternative.

Contraindications

- Patients who are hypersensitive to any component of this product;
- Patients with hepatic disease.
- Patients with hyperuricaemia and/or gouty arthritis.
- Safety for use in children has not been established.
- Nursing mothers. If use of this medicine is deemed essential, the patient should stop nursing.

Special warnings and precautions for use

AFT - Pyrazinamide Tablet 500mg should be used only when close daily observation of the patient is possible, and when laboratory facilities are available for performing frequent liver function tests and blood uric acid determinations. Pre-treatment examinations should include in vitro sensitivity tests of recent cultures of M. tuberculosis from the patients as measured against the usual anti-tuberculous medicines. Adverse effects for AFT - Pyrazinamide Tablet 500mg primarily involve the liver and vary from asymptomatic elevations of liver function tests to serious clinical manifestations of hepatic disease; therefore, liver function tests, especially alanine transaminase (ALT) and aspartate transaminase (AST) determinations, should be carried out prior to therapy, and then every two to four weeks during therapy. Therapy with AFT - Pyrazinamide Tablet 500mg should be withdrawn and not reinstated if signs of hepatocellular damage occur. Pre-treatment examinations should also include baseline serum uric acid determinations. AFT - Pyrazinamide Tablet 500mg inhibits renal excretion of urates, frequently resulting in hyperuricemia that is usually asymptomatic. If hyperuricaemia accompanied by an acute gouty arthritis occurs, therapy should be discontinued and not reinstated. AFT - Pyrazinamide Tablet 500mg should be used with caution in patients with a history of gout or diabetes mellitus, as their management may become more difficult. In the presence of renal insufficiency reduction in size and/or frequency of dose is recommended. Increased difficulty in controlling diabetes mellitus has been reported when diabetics are given pyrazinamide.

Use in Children

The safety of AFT - Pyrazinamide Tablet 500mg for use in children has not been established. Because of its potential toxicity, the use of AFT - Pyrazinamide Tablet 500mg in children should be avoided unless it is considered crucial.

Interaction with other medicines and other forms of interaction

Probenicid is known to block the excretion of pyrazinamide.

Fertility, pregnancy and lactation

There are no well-controlled studies in pregnant women. AFT - Pyrazinamide Tablet 500mg should be used during pregnancy only if the potential benefit justifies the potential risk to the foetus. Pyrazinamide is excreted in breast milk. If use of AFT - Pyrazinamide Tablet 500mg is deemed essential to a nursing mother, the patient should stop nursing.

Effects on ability to drive and use machines

There is no data to suggest that AFT - Pyrazinamide Tablet 500mg affects the ability to drive or use machines.

Undesirable effects

The most frequent adverse effect is hepatotoxicity. Transient increases in serum aminotransferase (transaminase) concentrations, jaundice, hepatitis and a syndrome of fever, anorexia, malaise, liver tenderness, hepatomegaly and splenomegaly have been

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reported. Rarely acute yellow atrophy of the liver and death have occurred. Hepatotoxicity appears to be dose related and may occur at any time during therapy. With a dose of 3g daily, hepatotoxicity occurs in approximately 15% of patients with jaundice occurring in 2-3%.

Pyrazinamide inhibits renal excretion of urates frequently resulting in hyperuricemia. This effect is usually asymptomatic but acute gout has occurred in some patients. Nongouty polyarthralgia reportedly occurs in up to 40% of patients. Urisuric agents administered concurrently may reduce pyrazinamide induced hyperuricemia. If hyperuricemia is severe or accompanied by acute gouty arthritis, treatment with pyrazinamide should be discontinued.

Mild arthralgia and myalgia have been reported frequently with pyrazinamide therapy. Maculopapular rash, fever, acne, porphyria, dysuria, interstitial nephritis and photosensitivity with reddish-brown discolouration of the exposed skin have been reported rarely.

Hypersensitivity reactions including rash, urticaria and pruritus have been reported. Gastrointestinal disturbances including aggravation of peptic ulcer, nausea, vomiting and anorexia have also occurred.

Thrombocytopenia and sideroblastic anaemia with erythroid hyperplasia, vacuolation of erythrocytes and increased serum iron concentrations have occurred rarely. Adverse effects on blood clotting mechanisms have also been reported rarely.

Reporting of suspected adverse reactions

Reporting suspected adverse reactions after authorisation of the medicine is important. It allows continued monitoring of the benefit/risk balance of the medicine. Healthcare professionals are asked to report any suspected adverse reactions <https://nzphvc.otago.ac.nz/reporting/>

Overdose

The stomach should be emptied by emesis or gastric lavage. Short acting barbiturates may be given for manifestations of CNS stimulation, analeptics for coma and artificial respiration and oxygen for respiratory failure. In the event of shock, a vasopressor agent, metaraminol bitartrate, should be given.

PHARMACOLOGICAL PROPERTIES

Pharmacodynamic properties

Pyrazinamide is a highly specific agent having a bactericidal effect on Mycobacterium tuberculosis but no activity against other mycobacteria or micro-organisms in vitro. Results of in vitro susceptibility testing with pyrazinamide are affected by the test media, inoculum size and pH. The MIC for M. tuberculosis is less than 20 µg/mL at pH 5.6 but it is almost completely inactive at a neutral pH. Pyrazinamide is effective against persisting tubercle bacilli within the acidic intracellular environment of the macrophages.

The exact mechanism of action has not been fully elucidated but appears to partly depend on conversion of pyrazinamide to pyrazinoic acid. Susceptible strains of Mycobacterium tuberculosis produce pyrazinamidase which is an enzyme that deaminates pyrazinamide to pyrazinoic acid. The in vitro susceptibility of a given strain of the organism appears to correspond to its pyrazinamidase activity.

Pharmacokinetic properties

Pyrazinamide is readily absorbed from the gastro-intestinal tract with peak serum concentration being reached about 2 hours after taking the dose. Plasma concentrations of pyrazinoic acid, which is the major metabolite, are generally greater than those of pyrazinamide and peak 4-8 hours after dosing.

Pyrazinamide is widely distributed in body tissues and fluids including the liver, lungs and CSF. It is not known of pyrazinamide crosses the placenta but it is excreted in breast milk.

Pyrazinamide is metabolised primarily in the liver by hydrolysis to pyrazinoic acid which is subsequently hydroxylated to the major excretory product 5-hydroxypyrazinoic acid. It is excreted via the kidney mainly by glomerular filtration. Approximately 70% of a dose appears in the urine within 24 hours (mainly as metabolites with 4-14% as pyrazinamide). The plasma half-life is 9-10 hours in patients with normal renal and hepatic function. The plasma half-life may be prolonged in patients with impaired renal or hepatic function.

Shelf life

2 years from the date of manufacture.

Special precautions for storage

Store below 30°C. Protect from light and moisture.

Nature and contents of container

Bottles of 100 tablets.



Manufacturer and Product Registration Holder:
Y.S.P. INDUSTRIES (M) SDN. BHD. (199001001034)
Lot 3, 5 & 7, Jalan P/7, Section 13,
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43000 Kajang, Selangor Darul Ehsan, Malaysia.

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